VLA WATER MASER OBSERVATIONS TOWARD HIGH MASS PROTOSTARS

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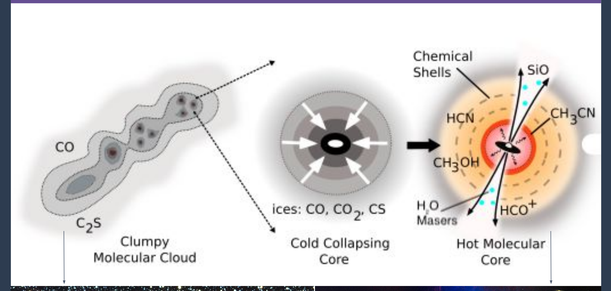
Introduction

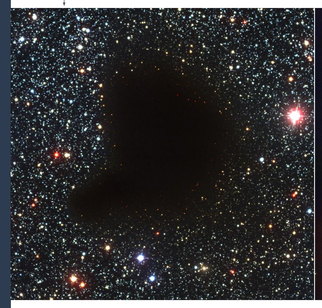
Astrophysical masers are regions of intense and coherent microwave radiation that can occur in various astronomical objects, such as stars, galaxies, and interstellar clouds. In the context of star formation, masers can be used as a powerful tool for detecting and studying the physical processes that occur during the early stages of star formation.

One of the main reasons why masers are a good source for detecting protostellar activity is that they are highly sensitive to the presence of dense and warm molecular gas, which is a key component of the protostellar environment. Masers are also capable of emitting radiation at very high brightness temperatures, making them detectable even at large distances.

In particular, H_2O masers have been extensively studied in the context of star formation, as they are commonly associated with young and embedded protostars. H_2O masers typically emit radiation at a frequency of 22 GHz and are characterized by their high brightness temperature, narrow linewidths, and high variability. They are thought to arise from the interaction between the outflows and disks surrounding protostars, and can provide valuable information about the kinematics and structure of these regions.

Background





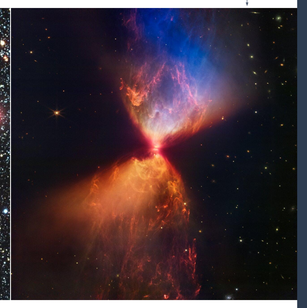


Fig. 1 (top) - A diagram depicting the stages of protostellar evolution from the Molecular Cloud phase till the Hot Molecular Core (HMC) phase with Masers in the paths of the ionised jets. Courtesy of Cormac Purcell.
Fig. 2 (left) - Picture of Barnard 68 (B68), a compact molecular cloud in the ISM. Courtesy of ESO.
Fig. 3 (right) - Image of Protostar L1527 in the Hot Molecular Core Phase with jets within the Taurus star-forming region. Courtesy of JWST.

Measurement & Methods

Fig. 4 - The Karl Jansky Very Large Array (VLA) at night. Courtesy of NRAO.

Data was taken from observations made using the Karl Jansky Very Large Array Telescope (VLA) as part of a proposal to study ionised jets from High-Mass Protostars.

- A-configuration - highest resolution and sensitivity (rms - 3μJy)
- Observations: 07 Mar to 06 Jun 2022
- Band: K-Band (18 - 26.5 GHz)
- Pointing Accuracy: 7"
- 23 high-mass star forming regions sampled
- Associated Continuum sources also sampled

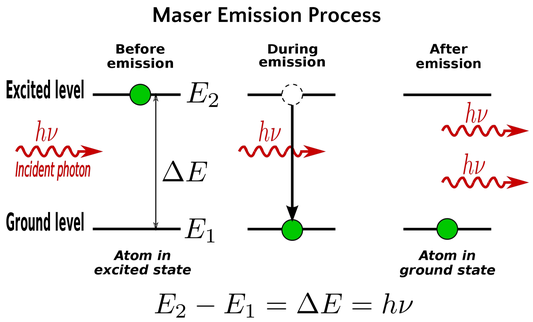


Fig. 5 - A diagram of the mechanism that causes masing - Energy Transitions from higher to lower states results in emissions that emit along the beams of the ionized jets.

Data

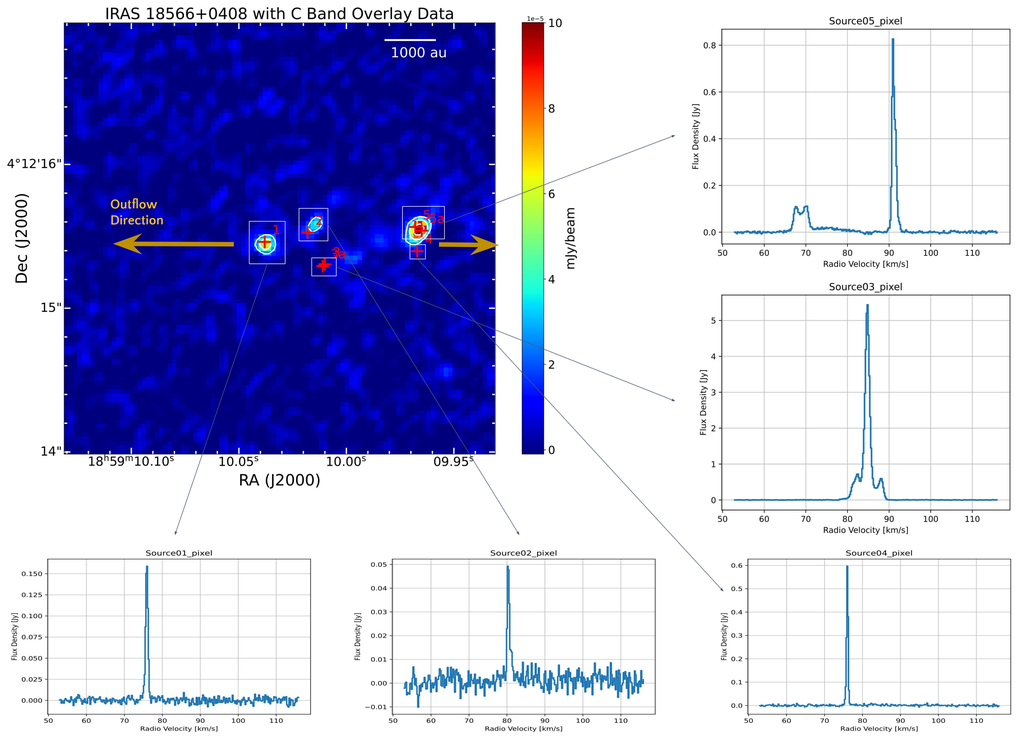


Fig. 6 - High-Mass Star Forming Region IRAS 18566+0408 with emission in the continuum, 5 maser sources (numbered 1 - 5a), their respective spectra, and a diagram of the observed direction of the molecular outflows.

Sources 3 and 5 have multiple observed emission peaks at slightly different centers and different channels, which is why we used a Beam size fitting instead of a peak pixel fitting.

Results

- Overall, we achieved a detection rate of 60.8%, with 14 of the 23 observed regions with H_2O masers emission present in them
- Results for one of our sources, IRAS 18566+0408 (Continuum Data with spectral plots above) are summarised in Table 1.
  - We detect continuum peaks in the VLA 1.3 cm band in the region
  - Source 3/3a has the largest maser emission at ~5.5 Jy/beam, while Source 2 has the smallest emission at ~50 mJy/beam
  - The H_2O (J=6-5) main isotopologue traces a bipolar molecular outflow in the E-W direction
- In summary, we believe that results of our preliminary analysis have been accurate when compared with historic data and the completion of this survey will present us with more complete information on the properties of these star-forming regions and their environments.

Table 1. Properties of the detected Maser Spots for IRAS 18566+0408

Maser	Peak Pos (RA)	Peak Pos (Dec)	I_ν (Jy beam^{-1})	V_{LSR}^{peak} (km s^{-1})	ΔV (km s^{-1})
(1)	(2)	(3)	(4)	(5)	(6)
1	18h59m10.037s	+04d12m15.460s	0.00252513	75.885	1.6851
2	18h59m10.017s	+04d12m15.525s	0.00078292	80.098	1.4745
3	18h59m10.010s	+04d12m15.291s	0.086353306	84.732	8.4256
3a	18h59m10.009s	+04d12m15.304s	0.079312862	84.732	8.4256
4	18h59m09.966s	+04d12m15.395s	0.009482214	75.885	1.0532
5	18h59m09.966s	+04d12m15.551s	0.013134142	90.840	5.4766 & 3.5809
5a	18h59m09.964s	+04d12m15.538s	0.004369742	90.840	5.4766 & 2.7383
6	18h59m09.821s	+04d12m18.840s	0.000308931	71.251	1.0532

References & Acknowledgements

- V. Rosero *et al* 2016 *ApJS* 227 25
- V. Rosero *et al* 2019 *ApJ* 880 99
- Tatiana M. Rodríguez *et al* 2021 *ApJ* 922 66
- Andrés E. Guzmán *et al* 2016 *ApJ* 826 208
- P. Hofner *et al* 2017 *ApJ* 843 99
- M. Elitzur et al 1989 ApJ 346 983

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